

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – III

Roll No.: 52	Name: Divyanjali yadav
Class: TYIT	Batch:
Date of Assignment: 14-07-2026	Date/Time of Submission: 14-07-2026

3(A).

Aim: Create methods add(), multiply(), subtract(), divide() with suitable parameters and call these methods using concept of C# delegate.

Algorithm:-

Step 1 :- Start the program.

Step 2 :- Create a delegate and a Calculator object.

Step 3 :- Assign each calculator method (Add, Sub, Mul, Div) to the delegate.

Step 4 :- Call the delegate with two numbers (40, 20).

Step 5 :- Display the results of all operations.

Step 6 :- End the program

Code: using System;

```

delegate void Operation(int a, int b);
class Calculator
{
    public void Add(int a, int b)
    {
        Console.WriteLine("Addition = " + (a + b));
    }
    public void Sub(int a, int b)
    {
        Console.WriteLine("Subtraction = " + (a - b));
    }
    public void Mul(int a, int b)
    {
        Console.WriteLine("Multiplication = " + (a * b));
    }
    public void Div(int a, int b)
    {
        Console.WriteLine("Division = " + (a / b));
    }
}
class Program
{

```

```

static void Main()
{
    Calculator c = new Calculator();
    Operation obj;

    obj = c.Add;
    obj(20, 10);

    obj = c.Sub;
    obj(20, 10);

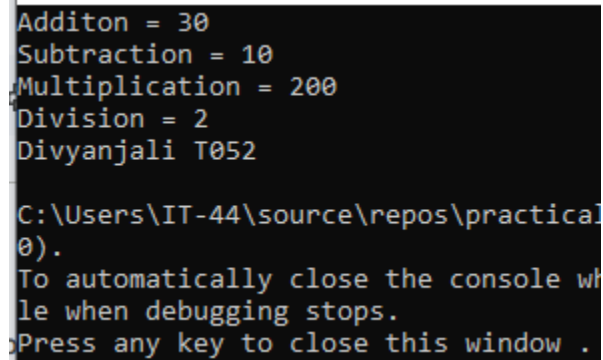
    obj = c.Mul;
    obj(20, 10);

    obj = c.Div;
    obj(20, 10);

    Console.WriteLine("Divyanjali T052");
}
}

```

Output:



```

Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2
Divyanjali T052

C:\Users\IT-44\source\repos\practical\bin\Debug\practical.exe
To automatically close the console window when debugging stops.
Press any key to close this window .

```

3(b).

Aim: Write a program using multicast delegates.

Algorithm:-

Step 1:- Start the program.

Step 2:- Create a delegate and object.

Step 3:- Add all methods to the delegate.

Step 4:- Call the delegate to display all messages.

Step 5:- Display the name.

Step 6:- End the program.

Code: using System;

```

delegate void Message();
class Demo
{
    public void Welcome()
    {

```

```

        Console.WriteLine("WELCOME...");
    }
    public void Learn()
    {
        Console.WriteLine("Learning C# Delegates");
    }
    public void Thanks()
    {
        Console.WriteLine("Thank You");
    }
}
public class Program
{
    public static void Main(string[] args)
    {
        Demo d = new Demo();
        Message m;

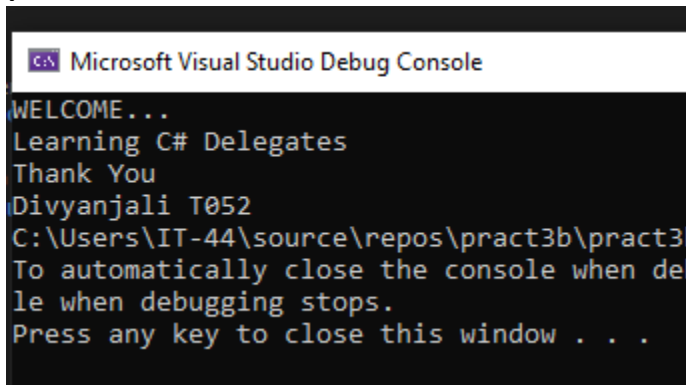
        m = d.Welcome;
        m += d.Learn;
        m += d.Thanks;
        m();

        Console.Write("Divyanjali T052");

    }
}

```

Output:



The screenshot shows the Microsoft Visual Studio Debug Console with the following output:

```

WELCOME...
Learning C# Delegates
Thank You
Divyanjali T052
C:\Users\IT-44\source\repos\pract3b\pract3b\
To automatically close the console when deb
le when debugging stops.
Press any key to close this window . . .

```

3(c).

Aim: Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Algorithm:-

Step 1:- Start the program.

Step 2:- Create a BankAccount object.

Step 3:- Set the account number and balance.

Step 4:- Display the account number and balance.

Step 5:- Update the balance and display the name.

Step 6:- End the program.

Code:

```

using System;
class BankAccount
{
    public int AccountNumber { get; set; }
    private double balance;

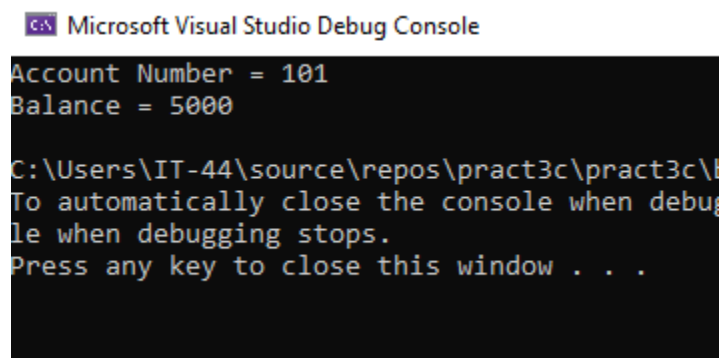
    public double Balance
    {
        get
        {
            return balance;
        }
        set
        {
            if (value >= 0)
                balance = value;
            else
                Console.WriteLine("balance cannot be negative");
        }
    }
}

class Prograqm
{
    static void Main()
    {
        BankAccount b = new BankAccount();

        b.AccountNumber = 101;
        b.Balance = 5000;
        Console.WriteLine("Account Number = " + b.AccountNumber);
        Console.WriteLine("Balance = " + b.Balance);

        b.Balance = 1000;
    }
}

```

Output:


Microsoft Visual Studio Debug Console

```

Account Number = 101
Balance = 5000

C:\Users\IT-44\source\repos\pract3c\pract3c\
To automatically close the console when debug
le when debugging stops.
Press any key to close this window . . .

```